B. Ignition Perspective vs. Edge Panel: Differences and Integration (8.1 / 8.3)

Appendix B of the EFC Gases & Advanced Materials Controls Standard, Revision H, October 2, 2026. Standalone copy; see the full document for the other chapters and appendices.

B.1 Overview

Perspective and Edge Panel are not competing products: Perspective is a visualization module, and Edge Panel is a license for a lightweight Edge gateway that can run Perspective (or Vision) with only two clients. The real comparison is a Standard Ignition gateway running Perspective versus an Edge gateway with the Panel product. The two integrate over the Gateway Network, mainly through a remote tag provider on the central server and Edge Sync Services for history, alarms, and audit data.

- **Ignition Standard + Perspective:** the full platform. Unlimited tags, sessions, and database connections, multiple projects and tag providers. This is the central SCADA server.
- **Ignition Edge Panel:** a local HMI at a machine or skid. It includes everything in Edge IIoT plus local visualization, limited to one project, no databases, and short local history. It keeps running if the link to the central server drops.

Mixed versions work with limits: an 8.3 main server can take tags, history, and alarms from an 8.1 Edge Panel, but an 8.3 Edge cannot store data to an 8.1 main server. Details are in the version sections below.

B.2 Key Differences

The biggest practical limits of Edge Panel are two clients, one project, one tag provider, and no database connections; everything else is close to the standard platform.

Table B.1: Standard Ignition vs. Edge Panel

Area	Standard Ignition + Perspective	Edge Panel (running Perspective)
Clients	Unlimited sessions (per license)	2 Perspective sessions, or 2 Vision clients (1 local, 1 remote); one module at a time
Projects	Unlimited, with inheritance	Exactly one project (default name “Edge”), cannot be deleted
Tags	Unlimited, multiple tag providers, remote providers	Unlimited tags, a single realtime tag provider
Databases	Unlimited SQL connections, named queries, DB user sources	None; DB queries and bindings do not work

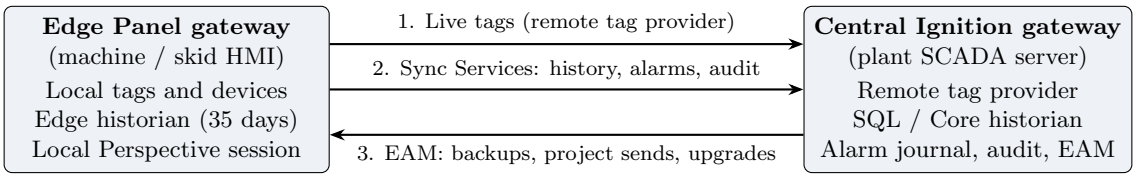
Table B.1 continued

Area	Standard Ignition + Perspective	Edge Panel (running Perspective)
Tag history	SQL Historian; 8.3 adds the Core Historian	Built-in Edge historian: 35 days or 10 million points, not configurable
Alarm journal / audit	Database-backed, long retention	7 days local each
Long-term storage	Native	Via Sync Services to a central gateway
Redundancy	Standard redundant pairs	Edge-to-Edge only; cannot fail over to a standard gateway
Scripting, WebDev, EAM agent, OPC UA server	Yes	Yes (bundled into Panel and IIoT since 8.1.33)
Third-party modules	Yes	Only if the vendor states Edge compatibility
Graphics	Perspective components, Symbol Factory add-on	Perspective or Vision, Symbol Factory included

Views built on a standard Perspective project can be exported and imported into the Edge project and vice versa, as long as they avoid database features. Sources: Ignition Edge (8.3), Edge Panel (8.3).

B.3 Integration Options

The standard pattern is a Gateway Network connection from the Edge Panel to the central server, then a remote tag provider for live data and Sync Services for history, alarms, and audit records.



The Edge Panel keeps the local HMI running if the network drops; the central gateway gives plant-wide views and long-term storage. All three links run over the Gateway Network.

Table B.2: Edge Panel Integration Methods

Method	What it does	Needs on central	Notes
Remote tag provider (Gateway Network)	Central Perspective views read and write Edge tags live	Nothing extra	Main integration path; write/edit access set in Edge security
Edge Sync Services	Pushes Edge tag history, alarm journal, and audit log to central storage	Historian, alarm journal, and audit profiles that allow remote storage	Edge buffers up to 35 days if the link is down
EAM (agent on Edge)	Central backups, project sends, upgrades, license management	Enterprise Administration module	Edge acts as an agent

Table B.2 continued

Method	What it does	Needs on central	Notes
Gateway messaging	Scripts call `system.util.sendMessage` / `sendRequest` between gateways	Message handlers in a central project	Good for events and recipes
Remote alarm notification	Edge alarms trigger central notification pipelines	Alarm Notification module	Uses the Remote Gateway Notification profile
MQTT (Sparkplug B)	Edge publishes tags to a broker; central subscribes	MQTT Engine module and a broker	Avoids Gateway Network version coupling
OPC UA	Central connects to the Edge's OPC UA server	OPC UA client connection	8.3 blocks anonymous writes by default

Shared screens: export views from the central Perspective project and import them into the single Edge project, or the reverse. Point bindings at the right tag provider on each side, and strip any database-backed features before importing to Edge.

B.4 Step-by-Step Setup

The procedure is the same on 8.1 and 8.3; mainly the gateway menu locations and security defaults change. 8.1 paths are approximate from the 8.1 manual layout, so confirm them on the gateway.

1. **Set Edge Panel to Perspective.** 8.1: Config > Gateway Settings > Edge Settings > Visualization Module. 8.3: Platform > System > Ignition Edge > Edge Settings tab. Switching faults the other module; restart the Designer after.

2. **Create the Gateway Network connection.** On the Edge, add an outgoing connection to the central server's address, port 8060 (SSL). Approve the certificate on the outgoing side, then approve the connection and certificate under Incoming Connections on the central server.

   - 8.3 defaults **Require Two-Way Authentication** to true, so both ends must trust each other.
   - Mixed 8.1/8.3: **Allow Java Serialization** must be enabled on every 8.3 gateway.

3. **Allow Edge storage on the central server.** 8.1: Config > Security > Service Security, set Query and Storage for History, Alarm Journal, and Audit on the Edge's zone. 8.3: Platform > Security > Zones > Manage Policy > Query and Storage for the same three sections.

4. **Configure Sync Services on the Edge.** 8.3: Platform > System > Ignition Edge > Sync Services. Pick the central gateway, then the remote history provider, alarm journal, and audit profile, and enable each. Set batch size, frequency, and an optional schedule.

5. **Create the remote tag provider on the central server.** 8.3: Services > Tags > Create Tag Provider > Remote Tag Provider (Gateway Network). 8.1: Config > Tags > Realtime > Create new Realtime Tag Provider > Remote Tag Provider. Choose the Edge gateway and its provider.

6. **Grant write access on the Edge.** In the Edge's security zone policy (Service Security in 8.1, Zones > Manage Policy in 8.3), set the Edge tag provider's access level to ReadWrite, or ReadWriteEdit to edit tags from a remote Designer.

7. **Build the central views.** Bind Perspective components on the central server to the Edge tag provider, for example `[EdgeProvider]Area/Tag`. Query synced history through the central historian,

not the Edge.

8. **Optional: EAM.** Install Enterprise Administration on the central server, make it the controller, and register the Edge as an agent for scheduled backups.
9. **Test failure modes.** Pull the network link and confirm the Edge HMI keeps running, the central server shows bad quality, and synced history back-fills when the link returns.

Source: Configuring Sync Services (8.3).

B.5 8.1 vs. 8.3 Differences

Edge Panel's limits are the same in both versions; the differences are in gateway security, serialization, historians, and menu layout.

Table B.3: Ignition 8.1 vs. 8.3 Differences Relevant to Edge Panel

Area	8.1	8.3
Edge Panel limits	2 clients, 1 project, 35-day history (since 8.1.34; 7 days before)	Same
Sync, EAM, Compute	Bundled into Panel and IIoT since 8.1.33	Bundled
Gateway Network serialization	Java serialization	Protobuf; Java serialization only allowed for 8.1 peers
Two-way authentication	Off by default	On by default; both ends approve certificates
Security settings	Service Security page	Moved into Security Zones > Manage Policy
Gateway web pages	Config / Status menus	Platform / Services / Connections layout; role-based access, 8.1 permissions not carried over
Central historian	SQL Tag Historian	SQL Historian plus new Core Historian (QuestDB); qualified path uses `sys:` and `prov:` instead of `drv:` for Core
History scripting	`system.tag.queryTagHistory` etc.	`system.historian.*` preferred; old functions still work
OPC UA	Anonymous clients can write by default	Anonymous clients read-only by default; new certificates on upgrade
Modules	Hot swap allowed	Restart required to install or upgrade a module
Launchers / Workstation	8.1 launchers	8.3 launchers work with 8.1 gateways, not the reverse
Perspective	—	Adds Drawing and Form components and Offline Mode; File Upload capped at 20 MB by default
Store and Forward	Max Records setting	Multi-threaded; renamed setting defaults to unlimited

If the Edge uses Perspective's newer 8.3 components, those views cannot be moved back to an 8.1 gateway. Source: 8.1 to 8.3 Upgrade Guide.

B.6 Mixed Versions: Can They Be Integrated?

Yes, with one direction that works and one that loses data: an 8.3 central server with 8.1 Edge gateways is supported, but an 8.1 central server cannot store data synced from an 8.3 Edge. Inductive Automation's rule is to upgrade the central gateway that hosts history, alarm journal, and audit first, then the Edge and remote gateways.

Table B.4: Mixed-Version Integration

Feature	Central 8.3, Edge 8.1	Central 8.1, Edge 8.3
Gateway Network link	Works; enable Allow Java Serialization on the 8.3 side	Works; enable Allow Java Serialization on the 8.3 side
Live tags (remote tag provider)	Read and write work	Read and write work
Editing remote tag configs from the other version's Designer	Not supported	Not supported
Sync Services (history, alarms, audit)	Works	Fails; data sits in the Edge buffer (35 days) then is lost
Remote history / alarm journal / audit	Works	Fails, same reason
EAM	8.3 controller manages 8.1 agents; some resources (Gateway and Perspective event scripts) cannot be sent to 8.1	Not recommended; upgrade the controller first
MQTT Sparkplug B	Works; each gateway needs Cirrus Link modules matched to its own version	Works, same condition
7.9 gateways	Cannot connect to any 8.3 gateway, even through a proxy	Same

Practical notes for a mixed period:

- Put the latest 8.1 patch on every 8.1 gateway before connecting it to 8.3; Inductive Automation strongly advises this before upgrading.
- An 8.3 gateway upgraded from 8.1 turns on Allow Java Serialization automatically; a fresh 8.3 install does not, so set it by hand.
- Plan to turn Allow Java Serialization off once every gateway is on 8.3, since 8.3-to-8.3 traffic uses Protobuf.
- Treat mixed versions as a transition state, not a long-term design; tag editing and project pushes across versions are limited.

For the planned 8.1 to 8.3 upgrade of the plant SCADA server, the order works in our favor: upgrade the central Perspective server first, confirm the Gateway Network shows the Edge as connected and sync is flowing, then upgrade the Edge Panel gateways.

Sources: 8.1 to 8.3 Upgrade Guide; 8.3 SDK Upgrade Guide.

B.7 Checklist

Use this list when commissioning an Edge Panel against the central server or moving through the upgrade.

- ☐ Gateway backups (`.gwbk`) of central and every Edge before changes
- ☐ Licenses confirmed 8.3-ready with the sales rep before upgrading
- ☐ Third-party modules confirmed compatible with 8.3, and with Edge where used
- ☐ Central server upgraded to 8.3 before any Edge gateway
- ☐ Allow Java Serialization on for every 8.3 gateway while any 8.1 gateway remains
- ☐ Certificates approved on both ends (two-way authentication)
- ☐ Security zone on central allows Query and Storage for history, alarm journal, and audit
- ☐ Sync Services enabled for all three data types and back-fill verified after a link outage
- ☐ Remote tag provider created; Edge write access set deliberately (ReadWrite only where operators need it)
- ☐ OPC UA write permissions re-mapped after 8.3 upgrade if any client connects anonymously
- ☐ Edge HMI tested with the network unplugged
- ☐ Allow Java Serialization turned off after the last gateway reaches 8.3

Sources

- Ignition Edge, 8.3 manual
- Edge Panel, 8.3 manual
- Configuring Sync Services, 8.3 manual
- 8.1 to 8.3 Upgrade Guide
- 8.1 to 8.3 SDK Upgrade Guide
- Edge Architectures, 8.1 manual